

CSA4302 IP
CO1 AT2

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1. HTTP Response Analysis

1. Incorrect/Invalid Field:

The Content-Length field is invalid because its value is - 150. Content length cannot be negative.

2. Issue with Content-Length:

Content-Length indicates the size of the response body in bytes. A negative value can cause incorrect processing of the response.

3. Status Code:

200 OK is correct if the request was successfully processed. However, the invalid Content-Length makes the response header incorrect.

4. Correction:

The Content-Length should contain a valid non-negative value representing the actual response size.

2. DOM Structure Analysis

1. Missing Element:

The <p> element is not properly closed. The correct structure should contain a closing </p> tag.

2. Browser Interpretation:

Browsers usually try to automatically correct improperly written HTML and create a valid DOM structure.

3. Impact:

Incorrect HTML may cause unexpected spacing, layout differences, or CSS and JavaScript issues.

4. Correction:

Properly close the <p> element and maintain correct nesting of all elements.

3. GET vs POST in Login System

1. Difference:

GET sends data through the URL, while POST sends data through the HTTP request body.

2. Suitable Method:

POST is more suitable for login because login credentials are not placed directly in the URL.

3. Security Risks of GET:

Passwords may be exposed through browser history, URLs, bookmarks, server logs, or shared links.

4. Best Practices:

Use POST with HTTPS, validate user input on the server, securely hash passwords, and use secure session management.

4. HTML Code Inspection

1. Missing Closing Tags:

The <h1> and <p> elements are missing their closing tags.

2. Issue:

The image should contain an alt attribute to describe the image and improve accessibility.

3. Browser Handling:

Browsers usually attempt to automatically correct missing tags, but this can result in unexpected DOM structures or rendering differences.

4. Correction:

Properly close the <h1> and <p> elements and add meaningful alt text to the image.

5. Benefit:

Correct HTML improves validity, accessibility, maintainability, and consistent rendering across browsers.

5. Browser-Server Communication Flow

1. Steps:

- The user enters a URL.**
- DNS converts the domain name into an IP address.**
- The browser sends an HTTP/HTTPS request.**
- The server processes the request.**
- The server sends an HTTP response.**
- The browser renders the received webpage.**

2. Possible Delays:

Delays can occur during DNS lookup, network communication, server processing, data transfer, and browser rendering.

3. Role of DNS:

DNS converts a domain name into its corresponding IP address so the browser can locate the correct server.

4. Performance Improvements:

Use DNS caching, CDN, compression, browser caching, optimized images, and faster server processing.